INCH-POUND

MIL-P-53103(ME)
7 February 1990

MILITARY SPECIFICATION

PUMP ASSEMBLIES, LOW TEMPERATURE, FLAMMABLE LIQUID, CENTRIFUGAL, SELF-PRIMING

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a low temperature, skid mounted, gas turbine-engine-driven, centrifugal pump assembly.

1.2 Classification. Pump assemblies shall be of the following types as specified (see 6.2).

Type I - Pump Assembly, low temperature, flammable liquid, 600 gpm (TA 13229E4000).
Type II - Pump assembly, low temperature, flammable liquid, 200 gpm (TA 13229E3900).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

GG-P-455 - Plates and Foils, Photographic.
QQ-S-781 - Strapping, Steel and Seals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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VV-F-800 - Fuel Oil, Diesel.
PPP-B-601 - Boxes, Wood, Cleated, Plywood.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-P-514 - Plates, Identification Instruction and Marking, Blank.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-G-3056 - Gasoline, Automotive, Combat.
MIL-T-5624 - Turbine Fuel, Aviation Grades JP-4 and JP-5.
MIL-L-7808 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-A-8625 - Anodic Coating, for Aluminum and Aluminum Alloys.
MIL-P-14105 - Paint, Heat-Resistant (for Steel Surfaces).
MIL-L-23699 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-T-83133 - Turbine Fuel; Aviation, Kerosene, Type JP-8.
MIL-L-85762 - Lighting, Aircraft, Interior, AN/AVS-t Aviator's Night Vision Image System (ANVIS) Compatible.

STANDARDS

FEDERAL

FED-STD-101 - Test Procedures for Packaging Materials.
FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down of Military Equipment.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-705 - Generator Sets, Engine-Driven, Methods of Tests and Instructions.
MIL-STD-810 - Environmental Test Methods and Engineering.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474 - Noise Limits for Army Materiel.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

TA13229E3900 - Pump Assembly, Low Temperature, Flammable Liquid, 200 GPM Capacity.
TA13229E4000 - Pump Assembly, Low Temperature, Flammable Liquid, 600 GPM Capacity.
TA13229E4150 - Tester Assembly, 200 GPM and 600 GPM Pump.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FSH, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 439 - Automotive Gasoline.
D 910 - Aviation Gasoline.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code - Steel.
D1.2 - Structural Welding Code - Aluminum.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJune Road, P. O. Box 351040, Miami, FL 33135.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC (SAE)

J429 - Mechanical and Material Requirements for Externally Threaded Fasteners.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The pump assembly shall consist of a gas turbine engine, electromagnetic clutch, and a centrifugal pump, together with all auxiliary equipment, systems, accessories, and controls assembled with protective housings and mounted on a skid base. A common engine and basic structure shall be used to interface either with the 200 gpm centrifugal pump and electromagnetic clutch or with the 600 gpm centrifugal pump and electromagnetic clutch. Conversion shall be accomplished by exchanging pump and clutch assemblies and applicable appurtenances. The 600 gpm pump assembly shall be in accordance with top assembly 13229E4000 and as specified herein. The 200 gpm pump assembly shall be in accordance with top assembly 13229E3900 and as specified herein. When specified (see 6.2), a tester assembly conforming to top assembly 13229E4150 shall be furnished with the pump assembly (see 6.1).

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those specified on the drawings to insure correct fit, assembly, interchangeability and operation of the pump assembly. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The pump and tester assembly shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the pump and tester assembly may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

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3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the pump and tester assembly may be newly fabricated from recovered materials to the maximum extent practicable, provided the pump and tester assembly produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the pump or tester assembly.

3.4 Pump assembly. The pump assembly shall conform to the drawings and as specified herein.

3.4.1 Interchangeability. Each part or component of the pump unit or tester assembly shall be identical in design and construction and be completely interchangeable with like parts or components on every other pump unit or tester assembly of the same purchase order or contract. Exceptions to this requirement must have the specific written approval of the contracting officer.

3.5 Maintainability. The pump assembly shall permit routine service and maintenance under military field conditions where personnel may be hampered by exposure to extreme environmental conditions. Components which require periodic servicing as a matter of routine maintenance shall be readily accessible. Covers or plates that are to be removed for operation or servicing shall be equipped with hinges and latches for easy access. Hinged covers shall be self-supporting and latch in the open position.

3.5.1 Maintenance support. Assemblages or support elements such as tester assembly, technical manuals, repair parts, special tools, or lubrication charts, as specified (see 6.5 and 6.6), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.5.2.8.

3.5.2 Maintenance ratio. The pump assembly shall have a maintenance ratio of not more than 0.07. Maintenance ratio is defined as a ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any test.

3.6 Transportability.

3.6.1 Rail transportability. Unless otherwise specified (see 6.2), the unpackaged pump assembly shall withstand rail impact without damage or permanent deformation to any part or component when tested as specified in 4.5.2.14. After testing, the pump assembly shall operate as specified herein for a period of not less than 1 hour without malfunction or permanent deformation of any part of the pump assembly.

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3.6.2 Road transportability. The pump assembly shall be capable of being transported in a convoy truck over roads as follows without malfunction, permanent deformation, or damage to the pump assembly or its components:

a. Over hard-surfaced roads at an average speed of 20 miles per hour (mph).
b. Over gravel roads at an average speed of 10 mph.
c. Over rough cross-country terrain at an average speed of 8 mph.

3.6.3 Slinging provisions. Permanently affixed slinging provisions conforming to MIL-STD-209 shall be provided that enable the pump assembly to be lifted in its normal travel position.

3.6.4 Tiedown provisions. Permanently affixed tiedown provisions conforming to MIL-STD-209 shall be provided that permit fastening the pump assembly to the float or deck of a transportation medium.

3.7 Performance characteristics.

3.7.1 Pumping capacity. The type II, 200 gpm pump assembly shall deliver a flow of 200 gpm of fuel having a specific gravity of 0.90 at a total head of not less than 375 feet. Type I, 600 gpm pump unit shall deliver a flow of 600 gpm of fuel having a specific gravity of 0.90 at a total head of 225 feet. Both the 200 gpm and 600 gpm pump assemblies shall be self-priming and shall prime within 2 minutes and shall be capable of delivering the above indicated output at all environmental conditions specified in 3.11 for the minimum time specified in 3.7.2 using the lubricants specified in 3.7.4 and the fuels specified in 3.7.3. The pump assembly shall be capable of pumping water at temperatures above 32 °F, aviation gasoline, automotive gasoline, JP-4, JP-5, JP-8 and diesel fuel.

3.7.2 Starting time. Each pump assembly shall be capable of starting, accelerating to rated speed, and accepting rated load within 120 seconds after initiating the starting cycle in ambient air temperatures -25 °F with fully charged batteries. In ambient temperature between -26 and -40 °F fully charged batteries or external power may be used. Below -40 °F warm, fully charged batteries or external power may be used.

3.7.3 Fuels. The pump assembly engine shall start and operate on all primary, secondary, and emergency fuels specified herein. The engine shall be capable of operating continuously without degradation on all primary and secondary fuels. There shall be no performance degradation when operating on emergency fuels for periods not to exceed 24 hours.

a. Primary fuel. VV-F-800.
b. Secondary fuels. MIL-T-5624 and MIL-T-83133.
c. Emergency fuels. MIL-G-3056, ASTM D 910, and ASTM D 439.

NOTE: The use of appropriate grades of fuels for existing ambient temperature is intended. The order of listing does not indicate order of preference for use.

3.7.4 Lubricating and cooling oils. Lubricating and cooling oils for the pump assembly shall be as follows:

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a. MIL-L-7808.
b. MIL-L-23699.

NOTE: The use of appropriate grades of these lubricants for existing ambient temperature is intended.

3.7.5 Tester assembly performance. The tester assembly shall simulate pump assembly performance utilizing three electronic logic packages (see 6.10). Two of the electronic packages shall have the printed circuit boards modified to reflect a known deficiency either high temperature or overspeed. The third electronic package shall have no deficiency in the printed circuit boards. An adapter cable shall be provided to interface the tester assembly with its power source PP-34/MSM battery charger to provide dc power for the tester assembly from a common available ac source.

3.7.6 Mobility aids. The pump assembly shall be provided with mobility aids (see 6.12). Transportability of the pump assembly would indicate that where feasible addition of mobilization devices should minimize the increase in outside dimensions. The mobility aids shall provide the pump assembly the capability to negotiate a ramp angle of 15 degrees without any part of the pump assembly dragging.

3.7.6.1 Pry bars. The pump assembly shall be provided with pry bars to loosen the pump assembly when ice or snow bound. The pry bars may also be used for lifting the pump assembly at the corners or at one end.

3.7.6.2 Sled bottom. A sled bottom fabricated of aluminum alloy shall be welded to the skid base and be designed to prevent the pump assembly from diving into snow or ice.

3.7.6.3 Towing eyes. Towing eyes shall be welded to the front and rear of each assembly for attaching a towing cable for tandem towing by the winch. A safe towing cable to interface with the pump assembly and filter/separator shall be provided.

3.7.6.4 Retractable/removable tires/wheel. The pump assembly shall be designed with low temperature, low ground pressure, removable/retractable wheels/tires.

3.7.6.5 Winch. An explosion proof electric winch shall be provided and located on the engine side of the chassis mounted on the frame. The frame and towing eyes shall be structurally strong enough to withstand forces of towing the pump assembly and a filter/separator (380 pounds dry weight) using the mobilization aids. The winch shall be capable of pulling on 45 percent grades. A 25 percent safety factor in determining capacity shall be considered essential to prevent winch damage during cold weather operation.

3.7.7 Second fuel source. The pump assembly shall have the capability to switch from the five gallon fuel can to the fuel stored in the system as an alternate fuel source to operate the gas turbine engine.

3.7.8 Army secure lighting requirement. All light sources shall meet the following secure lighting requirements:

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a. Light sources shall consist of incandecsnt bulbs with filters and dimmers or light emitting diodes (LEDs).

b. 95 percent of the light energy emitted by each light source shall be at wavelengths below 700 nanometers.
c. Incandescent light sources shall be dimmable to 0.05 foot-Lamberts (fL) or less.
d. All fault indicator and warning light sources shall be visible in bright sunlight.
e. Fault light and warning light indicators shall emit light in red/orange spectrums.

3.8 Human factors engineering. The pump or tester assembly shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, general requirements (4), control/display integration (5.1), visual displays (5.2), controls (5.4), labeling (5.5), anthropometry (5.6), design for maintainability (5.9), and hazards and safety (5.13) of MIL-STD-1472, as applicable.

3.9 Noise limits. The noise produced by the pump assembly shall conform with MIL-STD-1474 requirements, with the exception of MIL-STD-1474, 5.2, 5.3, and 5.4, when tested in accordance with 4.5.2.6. The provisions of MIL-STD-1474, 4.3 and 4.4, shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the contracting officer and written permission to exceed the 85 dB(A) limit is obtained from the contracting officer. Hazard signs shall conform with MIL-STD-1474, 4.3, except the sign will state "HEARING PROTECTION REQUIRED WITHIN (specify) FEET". The sign shall be readable at the distance indicated on the sign.

3.10 Safety. All shafts or rotating parts of the pump assembly shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the equipment, including exhaust pipes exposed to contact by personnel or which create a fire hazard, shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment. Adequate shielding may be used to attain the same purpose. Exhausts or discharges from the equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. Electrical equipment shall be effectively guarded or grounded to protect all persons and objects from electrical shock hazard.

3.11 Environmental requirements. The pump assembly shall be designed to withstand environmental conditions for operation and storage as specified below. The pump shall be operable by personnel when wearing arctic protective clothing or mission oriented protective posture (MOPP) IV chemical, biological and radiological clothing.

3.11.1 Operational requirements. The pump assembly shall start, operate, and meet all operating requirements as specified herein at any of the following conditions or combination of conditions. The pump shall not be required to pump water at temperatures below 32 °F.

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a. Temperature conditions.

(1) Any temperature from -60 to +60 °F.
(2) Any temperature from +60 to +100 °F with not more than a 10 percent loss in flow capacity.

b. Pressure conditions. Any altitude from sea level (29.92 in. Hg) up to 8000 feet (22.20 in. Hg) at a maximum temperature of 60 °F, must meet 90 percent of all operating requirements stated above (flow capacity) at altitudes of up to 5000 feet (24.78 in. Hg) within the temperature range of +60 to +100 °F and at altitudes above 5000 feet (24.78 in. Hg) at a temperature of +60 to +100 °F maximum.

c. Precipitation. During rainfall as specified in MIL-STD-705, method 711.3b.

d. Relative humidity. As low as 5 percent at a temperature of 100 °F and as high as 100 percent at 100 °F.

e. Wind. With wind up to 45 knots for a 5-minute period, with gusts to 65 knots, and, using holddowns, with winds of 55 knots for 5-minute periods with gusts to 85 knots.

3.11.2 Storage requirements. The pump assembly shall not be damaged by storage in temperatures from +165 to -65 °F.

3.12 Reliability. The specified mean-time-between-failure (MTBF) shall be 300 hours when the pump assembly is operated at rated capacity and tested as specified in 4.5.2.7.

3.13 Electromagnetic interference. The electromagnetic interference and susceptibility characteristics shall not exceed the UM03 limits for class C1 equipment of MIL-STD-461.

3.14 Weight. The overall weight for the type I pump assembly shall not exceed 795 pounds, and for the type II, 685 pounds excluding the addition of mobilization aids.

3.15 Treatment and painting. Except as otherwise specified (see 6.2), the pump assembly shall be treated and painted in accordance with MIL-T-704, type F or G, color Green 383, conforming to MIL-C-46168. Treatment and painting of components interior to the pump assembly may follow the manufacturer's standard practice if such practice is compatible with the environmental and other performance requirements specified herein, except that interior parts (panels, brackets, angles, channels, etc.) of the housing shall be treated in the same manner as external surfaces. Stainless steel parts need not be painted, except that the exhaust box shall be painted with heat-resisting paint conforming to MIL-P-14105. The following items shall not be painted: electrical terminals and connections, instruction plates, nameplates, switches, sound-insulating materials, and all other parts whose operation would be adversely affected by paint.

3.16 Identification plates. Identification plates shall be in accordance with MIL-P-514, type III, composition A, class 2, smooth surface, and shall be anodized in accordance with MIL-STD-8625, type 2, class 1. Plates having a thickness in excess of 0.050 inch shall be anodized after all machining has been completed. Plate background shall be black anodized having color in accordance with FED-STD-595, color no. 27038. Plate copy shall be clear anodized. The

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finished plates shall pass all tests in accordance with GG-P-455, grade A, class 1, except that the temperature of the oven shall be 400 °F and the time of exposure shall be 3 hours during the heat resistance test.

3.16.1 Nameplates and product marking. The pump and tester assembly shall be identified and marked and have information and instruction plates as specified herein. Plates shall be made of anodized aluminum not less than 0.030-inch thick permanently affixed to the set with rivets or bolts and processed as specified in 3.16.

3.16.1.1 Unit identification plate. A plate containing the following information shall be mounted on a conspicuous structural member of the pump and tester assembly. Federal stock number, technical manual number, serial number, manufacturer's name and date of manufacture, contract number, length, width, height, weight (dry), engine manufacturer, and model number. All applicable ratings of the pump assembly (capacity, hydraulic head, and rpm) shall be shown on the plate. The plate shall have a place for the Government inspector's stamp and date of inspection.

3.17 Identification marking. The parts of the pump assembly shall be identified in accordance with MIL-STD-130.

3.18 Instruction plates, diagrams, or labels. Each pump assembly shall be equipped with instruction plates, diagrams, or labels including warnings and cautions, describing any special or important procedure to be followed in assembling, operating, and servicing the pump assembly. Instruction plates and diagrams shall be as specified in 3.16 and labeling requirements shall conform to MIL-STD-1472. Plates, diagrams, and labels shall be attached by screws, bolts, or rivets in a location where the plate, diagram, or label will be both visible and legible.

3.18.1 Operating instructions. A plate containing operating instructions shall be mounted in a conspicuous location. The operating instructions shall be complete and brief; shall describe procedures for starting, operating, and stopping at ambient temperatures from +95 to -60 °F.

3.19 Workmanship. All details of workmanship shall be of the highest grade consistent with the intention of this specification. Particular attention shall be given to soldering, wiring, welding, brazing, plating, riveting, finishes, machine operations, and marking. There shall be no cracks, dents, scratches, burrs, sharp edges, loose parts, foreign matter, or any other evidence of poor workmanship.

3.19.1 Castings and forgings. All parts, components, and assemblies of the pump assembly which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.19.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of materials shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to

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insure uniformity of size and shape. Precautions shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free from burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.19.3 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than two full threads.

3.19.4 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for materials joined and type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 - Structural Welding Code - Steel.
AWS D1.2 - Structural Welding Code - Aluminum.
ASME Boiler and Pressure Vessel Code, Section IX.

Contractors who only make horizontal welds need not qualify welders for "all position welding". The contractor is responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator. The test results shall be made available for review by the contracting officer or designated representative.

3.19.4.1 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mil scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the pump or tester assembly to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 1/16 inch unless appropriate bridging techniques are used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The

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absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.1.3 Disassembly inspection. Failure of any examination or test by the first article shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and reference drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first articles. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.4 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.7 and 6.8.1).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection. The first article inspection shall be performed by the contractor. Acceptance of a first article pump and tester assembly shall not exclude the remaining pump and tester assemblies from the quality conformance inspection and acceptance provisions specified in section 4. In addition to any inspection specified as part of the first article inspection, the Government reserves the right to conduct any and all inspections contained in this specification as part of the first article inspection and failure of such additional inspections shall have the same effect as failure of those inspections specified as first article inspection. Rejection of any first article pump and tester assembly shall be cause for refusal by the Government to continue acceptance of production pump and tester assemblies until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on pump and tester assemblies previously accepted and produced under the contract. Any deficiencies found as a result of the first article inspection will be considered prima facie evidence that all pump and tester assemblies accepted prior to the completion of first article

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inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.3.1 Rejection and retest. When a pump or tester assembly fails to meet any of the requirements of the first article test, the cause of failure shall be determined. Full particulars of the failure and the actions required to correct the failure shall be reported to the contracting officer. All first article tests which are affected by the corrections shall be repeated.

4.3.2 Examination. The pump and tester assembly shall be examined as specified in table I. Presence of one or more defects shall be cause for rejection.

4.3.3 Tests. The first article pump assembly and tester shall be subjected to tests marked "x" in column 1 of table II. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I or table IIa. A lot shall be accepted when 0 defects are found and rejected when 1 or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests.

4.4.3.1 Individual. Each pump and tester assembly shall be subjected to the tests marked "x" in column 2 of table II. Failure of any test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "x" in column 3 of table II. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The pump and tester assemblies shall be examined as specified herein for the defects listed in table I.

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TABLE I. Examinations.

Examination No.	Examination description	Requirement Paragraph
101.	Materials not as specified.	3.3
102.	Materials not resistant to corrosion and deterioration or treated to be made resistant to corrosion and deterioration	3.3.1
103.	Dissimilar metals not in accordance with MIL-STD-889.	
104.	Material and Material finish or treatment information not available.	3.3.3
105.	Recovered materials not as specified.	3.3.4
106.	Pump or tester assembly not as specified on drawings.	3.4
107.	Interchangeability not as specified.	3.4.1
108.	Slinging and tiedown not as specified.	3.6.3 & 3.6.4
109.	Human factors engineering not as specified.	3.8
110.	Noise control system not as specified.	3.9
111.	Welding not as specified.	3.19.4.1
112.	Threads and threaded connections not as specified.	3.19.3
113.	Parts interchangeability not as specified.	3.4.2
114.	Maintenance support manual not as specified.	3.5.1
115.	Mobility aids not as specified.	3.7.6
116.	Weight not as specified.	3.14
117.	Treatment and painting not as specified.	3.15
118.	Identification plates not as specified.	3.16
119.	Identification marking not as specified.	3.17
120.	Instruction plates, diagrams, or labels not as specified.	3.18
121.	Workmanship not as specified.	3.19

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, test shall be performed in accordance with the applicable provisions of test code standards for centrifugal pumps of the Hydraulic Institute. All performance data shall be corrected to standard sea level conditions (barometric pressure of 29.92 inches of mercury and liquid fuel at 60 °F, having a specific gravity of 0.78). Petroleum fuels or a suitable solvent may be used as the test fluid. The test fluid temperature shall be the same as the ambient operating test temperature.

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4.5.2.2 Test schedule.

TABLE II. Test schedule.

First Article	Quality Conformance		Test	Test Method	Requirement Paragraph
	Individual	Sample			
1	2	3	4	5	6
x			Maintainability.	4.5.2.8	3.5
x			Rail impact.	4.5.2.14	3.6.1
x			Road transportability.	4.5.2.11	3.6.2
x			Slinging provisions.	4.5.2.12	3.6.3
x			Tiedown provisions.	4.5.2.13	3.6.4
x			Performance:		
x			Mobility aids.	4.5.2.3	3.7.6
x			Pump.	4.5.2.9.1	3.7.1.3.4
x			Priming.	4.5.2.9.2	3.7.1.3.4
x	x	x	Operational.	4.5.2.9.3	3.7.1.3.4
x		x	Production.	4.5.2.9.4	3.7.1.3.4
x			Altitude operation 1/	720.1	3.11.1
x			Electromagnetic interference.	4.5.2.10	3.13
x			Low temperature. 2/		3.7.2
			Starting time (-60 °F, -50 °F, -40 °F, -30 °F, -20 °F, -10 °F, 0 °F, +10 °F, +30 °F, +40 °F, +50 °F, +60 °F, and +100 °F	Procedure II	
x	x		Tester assembly.	4.5.2.5	3.7.5
x			Noise level measurement.	4.5.2.6	3.9
x			Safety.	4.5.2.4	3.10
			Environmental:		
x			Low temperature:	4.5.2.3.2	3.7.4, 3.11
x			Storage	4.5.2.3.2	3.11.2
x			Operation	4.5.2.3.2	3.11.1
x			High temperature:	4.5.2.3.1	3.11.2
			Storage	4.5.2.3.1	3.11.2
			Operation.	4.5.2.3.1	3.11.1
x			Reliability.	4.5.2.7	3.12
x			Second fuel source	4.5.2.15	3.7.7
x			Army secure lighting	4.5.2.16	3.7.8

1/ MIL-STD-705
2/ MIL-STD-810

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4.5.2.3 Environmental.

4.5.2.3.1 High temperature. The pump assembly shall be tested as specified in MIL-STD-810, procedures I and II, method 501. The maximum ambient storage temperature shall be 165 °F and the maximum ambient operating temperature shall be 100 °F. The operating period of procedure II shall be 2 hours. The pump assembly shall be operated at rated capacity and the performance of each mobility aid shall be tested. There shall be 200 starts and stops at the ambient temperature range of 70 to 100 °F with at least 50 starts and stops at 90 to 100 °F. Nonconformance to 3.7.6 and 3.11.2 shall constitute failure of this test.

4.5.2.3.2 Low temperature. The pump assembly shall be tested as specified in MIL-STD-810, procedures I and II, method 502. The minimum ambient storage temperature shall be -65 °F. The minimum ambient operating temperature shall be -60 °F. The operating period of procedure II shall be 2 hours. The pump assembly shall be operated at rated capacity using the lubricants specified in 3.7.4 and the fuels specified in 3.7.3. The performance of each mobility aid shall be tested at each temperature range specified herein. There shall be 300 start and stop cycles at the ambient temperature range of -60 to +69 °F; at least 50 start and stop cycles at each of the following: from -60 to -50 °F; from -49 to -40 °F; from -39 to -20 °F; and from -19 to 0 °F; and at least 100 start and stop cycles at ranges from +1 to +69 °F. Nonconformance to 3.7.6, 3.11, and 3.11.1 shall constitute failure of this test.

4.5.2.4 Safety. The pump assembly shall be evaluated for safety requirements throughout testing. Nonconformance to 3.10 shall constitute failure of this test.

4.5.2.5 Tester assembly. The tester assembly performance shall be tested utilizing the three electronic logic packages specified in 3.7.5. The tester assembly shall check simulation of all pump operations while utilizing each of the electronic packages. Failure to detect the two known deficiencies or appearances of a deficiency in the perfect electronic package shall constitute failure of this test.

4.5.2.6 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4; contours shall be taken at not fewer than 12 equal (horizontal) arc increments, one increment shall include data from the noisiest position; in addition, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.7 Reliability. Using the MTBF specified in 3.12, at least three and not more than five pumping assemblies shall be tested as specified in 4.5.2.9 for 780 hours with one pump accumulating at least 300 hours. Accept if 5 or less failures occur and reject if 6 or more failures occur. A failure is defined as any malfunction which cannot be corrected within 30 minutes (all waiting periods are excluded) by adjustment, repair, or replacement action using controls and on-equipment tools or parts available to the using element, and which causes or may cause one or more of the following:

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a. Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
b. Damage to the pumping assembly by continued operation.
c. Safety hazards to personnel.

Nonconformance to 3.12 shall constitute failure of this test.

4.5.2.8 Maintainability.

4.5.2.8.1 Maintainability test. The maintainability test shall be conducted concurrently with the reliability test (see 4.5.2.7). Data collected during maintenance activities shall be used to determine compliance with maintainability requirements in 3.5.1.

4.5.2.8.2 Maintenance support evaluation. Maintenance operations required during first article testing shall be accomplished to determine conformance to 3.5. Failure of the support elements to maintain the pump assembly as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the MTBF.

4.5.2.8.3 Maintenance ratio. The maintenance ratio shall be computed during the first article testing. Nonconformance to 3.5.2 shall constitute failure of this test.

4.5.2.9 Performance.

4.5.2.9.1 Pump. With the pump operating at 8000 ±500 rpm and a static suction lift equivalent to 10 feet of water, test the pump in accordance with the Rating Standards and Test Code Standards for centrifugal pumps of the Hydraulic Institute. Data relative to head-capacity shall be taken at a minimum of 6 points and shall include the rating point, wide open discharge, and shutoff. The type I and type II shall deliver not less than the capacity specified in 3.7.1.

4.5.2.9.2 Priming. Fill the pump case with either liquid petroleum fuel or a suitable solvent having a specific gravity of 0.72 to 0.75 and conduct the priming test with the pump operating at 8000 ±500 rpm and a static suction lift equivalent to 10 feet of water at standard atmospheric conditions. Failure of the pump to prime in not more than 2 minutes shall constitute failure of this test.

4.5.2.9.3 Operational. Fill the pump case with test fluid and prime the pump assembly with a static suction lift equivalent to 10 feet of water at standard atmospheric conditions. Once the pump has primed, operate the pump at the capacity specified in 3.7.1 for 100 hours. This test shall be performed for not less than 9 continuous hours daily until the conclusion of the test. During operation, only minor adjustments to the engine may be performed. A maximum shutdown time of 30 minutes is allowable for engine maintenance and examination during each 9-hour period. During operation, determinations shall be made as to whether there is leakage through the pump case, piping, fittings, or at the pump shaft seal and whether there is malfunction of any component. Leakage, malfunction of any component, inability of the pump assembly to prime in 2 minutes and deliver the capacity specified, or inability of the pump assembly to operate continuously during periods of operation shall constitute failure of this test.

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4.5.2.9.4 Production. Test the pump assembly as specified in 4.5.2.9.3 except that the operating time shall be 15 minutes.

4.5.2.10 Electromagnetic interference. The electromagnetic interference test shall be performed in accordance with MIL-STD-462, procedures RE02 and RS03. The equipment shall be considered to be a class C1, group II item. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.5.2.11 Road transportability. The first article pump assembly shall be hauled over a smooth, hard-surfaced road at an average speed of 20 mph for a total of 100 miles over a gravel road, at an average speed of 10 mph for 100 miles, and cross-country at an average speed of 8 mph for 100 miles. After each of the above segments of the test, the pump assembly shall be operated at rated capacity for 1 hour. Evidence of malfunction, permanent deformation, or damage to the pump assembly components, or failure of the pump assembly to operate at rated capacity shall constitute failure of this test.

4.5.2.12 Sling provisions.

4.5.2.12.1 Test procedure. Restrain the pump assembly by anchoring the main frame and subject the lifting bail to a lifting force of 2.5 times the weight of the pump assembly. An acceptable alternate method of test shall be to lift the pump assembly clear of the ground by lifting bail and add weights to the pump assembly until the force of the lifting bail is 2.5 times the weight of the pump assembly. Hold the pump assembly under either of the above loading conditions for not less than 90 seconds.

4.5.2.12.2 Failure criteria. Weld failure, permanent deformation of the pump assembly or the lifting bail, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.5.2.13 Tiedown provisions. With the pump assembly anchored by means other than the tiedown being tested, apply the static loads specified in MIL-STD-209 to each provision. Measure the dimensions of the tiedowns. Weld failure, permanent deformation of the pump assembly or its tiedown provisions, breakage of the provision, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.5.2.14 Railroad car impact. Anchor the unpackaged pump assembly on a domestic service railway freight car (blocking is permitted). Shunt the car, with the pump assembly tied down, at speeds of not less than 4, 6, and 8 mph against two stationary flatcars having a total gross weight of at least 240,000 pounds with their airbrakes set in the emergency application position and draft gear slack removed. The pump assembly shall face in the direction of the two cars. Repeat the procedure at not less than 8 mph with the rear of the pumping assembly in the direction of the two cars. The pump assembly may be reversed by either reversing the direction of the flatcar (preferred) or by reloading the pump assembly in the proper direction on the freight car. Once the test has begun, there shall be no readjustment of load nor any reconditioning of the blocks, choking material, or tiedowns, except as necessary to reverse the direction of the pump assembly. After each impact, the pump assembly shall be inspected. After completion of the railroad car impact test, an operational

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checkout of the pump assembly shall be conducted. Nonconformance to 3.6.1 shall constitute failure of this test.

4.5.2.15 Second fuel source. While conducting pumping operations the pump assembly shall continue to operate as the engine fuel source is switched from a five gallon fuel can to the fuel stored and being pumped in the system; after operating for 15 minutes switch from the fuel stored and being pumped to the fuel in a five gallon fuel can.

4.5.2.16 Army secure lighting. Testing of light sources with dimmers and filters shall be in accordance with MIL-L-85762, table V.

4.6 Inspection comparison. The Government may select pump assemblies at any time during the contract production period and subject these pump assemblies to the examination and tests specified in 4.3 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on pumping assemblies selected at random from those which have been accepted by the Government and will not include the previously inspected first article pump assemblies. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other test contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison pump assembly to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison pump assembly and shall be cause for refusal by the Government to continue acceptance of production pump assemblies until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on pump assemblies previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all pump assemblies accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging.

4.7.1 First article pack inspection.

4.7.1.1 Examination. The first article pack shall be examined for the defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.1.2 Test. The first article pack shall be tested in accordance with the pendulum-impact test, FED-STD-101, method 5012. Damage to the contents or the shipping container, rendering either useless for its intended use, shall constitute failure of the pack.

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4.7.2 Quality conformance inspection of packaging.

4.7.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

122.	Pumps not preserved in accordance with the reference document for level A (see 5.2.1).	
123.	Pumps not preserved as specified for level C (see 5.2.2).	
124.	Shipping container not as specified for levels A or C (see 5.3.1 and 5.3.2).	
125.	Strapping not as specified for levels A or C (see 5.3.1 and 5.3.2).	
126.	Marking missing, illegible, incorrect, or incomplete for levels A or C (see 5.4).	

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination and test within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The first article pack may be accomplished utilizing either the first produced pump assembly or a production pump assembly. If the first produced pump assembly is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the first produced pump assembly and the production pump assembly.

5.2 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.2.1 Level A. Each pump shall be preserved in accordance with the applicable requirements of submethod IIA of MIL-P-116.

5.2.2 Level C. Each pump shall be filled with the proper operating oil and covered with a waterproof barrier shroud. The shroud shall be secured at the base of the pump.

5.3 Packing. Packing shall be level A or C as specified (see 6.2).

5.3.1 Level A. Each pump shall be packed in a box conforming to PPP-B-601, overseas type, style A, modified as described herein. The pump shall be secured to the container in accordance with MIL-STD-1186. The containers shall be designed to be reusable by assembling the sides, end, and top as a reusable box cover. Battens, 2 x 2 inches in cross section, shall be positioned on the upper surface of the box base to provide a means to properly position the cover.

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Battens shall be located a distance from the edges of the box base equal to the thickness of the cleat and plywood of each side and end panel. The cover shall be secured to the container base with strapping conforming to QQ-S-781, class 2, type I or IV, finish B, size as applicable. Strapping shall encircle the box girthwise both laterally and longitudinally; those straps perpendicular to the skids shall be placed between the skids and the lower surface of the box bottom panel.

5.3.2 Level C. Each pump shall be packed as specified for level A except the box shall be domestic and the strapping shall be finish A.

5.4 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The type I pump assembly covered by this specification is a component of the Arctic Fuel System Supply Point: 60,000 to 120,000 gallon capacity, and is intended to be used for severe low temperature transfer and dispensing of liquid fuel. The type II pump assembly is a component of the Arctic Forward Area Refueling Equipment and is intended to be used for severe low temperature refueling of tactical vehicles and aircraft in a forward area. The tester assembly is used primarily to test for malfunction of either the 600 gpm or 200 gpm pump assembly printed circuit boards. The pump assembly printed circuit boards control starting, acceleration, operation and emergency shutdown of the pump assembly.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Type required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a tester assembly shall be furnished with a pump assembly (see 3.1).
e. Time frame required for submission of first article (see 3.2).
f. When the Government will conduct any or all of the first article examinations and tests. When the Government will conduct some but not all of the first article examinations and tests, the contracting officer should specify which examinations and tests will be conducted by the Government and which examinations and tests will be conducted by the contractor (see 3.2 and 4.3).
g. When rail impact testing is not required (see 3.6.1).
h. Paint color required when other than as specified (see 3.15).
i. Time frame required for submission of first article pack (see 5.1).
j. Level of preservation and packing required (see 5.2 and 5.3).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the

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first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the pump assemblies in accordance with this specification.

6.5 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools, as necessary (including any special tools), and instructions regarding shipment of pump or tester assembly.

6.7 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.8 Definitions. The following definitions apply to the text of this specification.

6.8.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. A QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.8.2 Start. A pump unit is considered to have started when the engine has accelerated to full governed speed and is operating at rated conditions.

6.8.3 Temperature and humidity damage. Temperature and humidity damage is defined as corrosion, breakage, deformation, reduction of insulation resistance below 50,000 ohms, or conditions causing malfunction of any component part.

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6.8.4 Dangerous flexural vibration. Dangerous flexural vibration is defined as a vibration which occurs at a speed at which maximum stress in the shaft from flexural vibration exceeds 9,000 psi.

6.8.5 Dangerous torsional critical speed. Dangerous torsional critical speed is defined as the speed at which maximum vibrating stress in the shaft from torsional vibration exceeds 5,000 psi.

6.8.6 Rough handling damage. Rough handling damage is defined as deformation, loosening, breakage, change in fit of any component or part, or any condition resulting in malfunction of the unit.

6.8.7 Normal railroad transportation. Normal railroad transportation will be interpreted to mean impact speeds up to and including 8 miles per hour under the test conditions specified in MIL-STD-705, method 740.5.

6.8.8 Normal aircraft and helicopter transportation. Normal aircraft and helicopter transportation will be interpreted to mean a 9-inch free fall drop and an 9-inch edge drop under the test conditions specified in MIL-STD-705, methods 740.2 and 740.3.

6.8.9 Weight. The weight of the unit is a dry weight and does not include the weight of fuel, lubricating oil, electrolyte, hydraulic oil, and optional equipment.

6.8.10 Dripproof enclosure. A dripproof enclosure is an enclosure so constructed that falling drops of liquid or solid particles striking the enclosure at an angle from 0 to 15 degrees from the vertical cannot enter the enclosure either directly or by striking and running along a horizontal or inwardly inclined surface.

6.8.11 Meter accuracy. Accuracy is a number which defines the limit of error expressed as a percentage of full-scale value. Error is the difference between the indication and the true value of the quality measured. It is the quality which, when algebraically subtracted from the indication, gives the true value. A positive error denotes that the indication of the meter is greater than the true value.

6.8.12 Fail-safe. A circuit arrangement such that a malfunction anywhere in the circuit (broken wire, operation of protective device, etc.) will shut down the unit and prevent the unit from being started.

6.8.13 Failure. A "relevant failure" is defined as any malfunction which the operator cannot remedy within 30 minutes by normal adjustment action using the unit controls and OEM equipment and which causes or may cause any or all of the following: Inability to commence operation; cessation of operation or degradation of performance capability of the system/subsystem below designated levels; serious damage to system/subsystem by continued operation; or create serious personnal hazard. Simultaneous, related malfunctions are considered one failure. Consideration must be made for relevancy of failure definitions and can be routinely corrected during user organization schedule maintenance.

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6.8.13.1 Random failure. Any failure whose occurrence is unpredictable in an absolute sense but which is predictable only in a probabilistic or statistical sense.

6.8.13.2 Irrelevant failure. Any failure which is not used to compute set reliability.

6.8.14 Availability.

6.8.14.1 Inherent availability.

$$\text{Inherent availability (Ai)} = \frac{MTBF}{MTBF + MTTR}$$

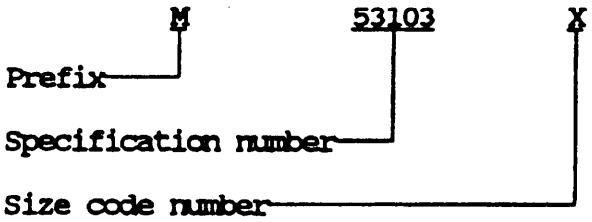
Where: MTBF = Mean time between failure.
MTTR = Mean time to repair.

6.8.14.2 Achieved availability.

$$\text{Achieved availability (Aa)} = \frac{MTBM}{MTBM + M}$$

Where: MTBM = Mean time between maintenance
M = Mean active maintenance downtime resulting from both preventive and corrective actions.

6.9 Part or identifying number (PIN). The PIN to be used for pump assemblies acquired to this specification are created as follows:



6.10 Subject term (key word) listing.

Pump assembly, skid mounted.
Pump assembly, turbine-engine-driven.

6.11 Electronic logic package. The electronic logic package is an aluminum box 10 x 6 x 4 inches that contains printed circuit boards that control starting, acceleration, operation and emergency shutdown of the pump assembly.

6.12 Mobility aids. The contracting officer should specify that complete design drawings be provided to fully describe the mobility aids and interfaces or any resulting pump assembly design modifications.

6.13 Mission oriented protective posture (MOPP). MOPP IV protection consists of a two piece protective overgarment, hood, overboots, and rubber gloves with cotton liners. The overgarment is closed and hood is pulled down and adjusted.

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Army - ME

Project 4320-A279

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-53103(ME)	900207

3. DOCUMENT TITLE

Pump Assemblies, Low Temperature, Flammable Liquid, Centrifugal, Self-Priming

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	c.
d. ADDRESS (Include Zip Code)	e. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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